

Project Composite: Wyckoff Institutional Bloodhound

1. Project Overview

Target Market: National Stock Exchange of India (NSE) **Core Utility:** Anomaly-First (Backward) Stock Discovery Pipeline focusing exclusively on Wyckoff Stage 1 Accumulation Floor Detection. **Core Philosophy:** Markets are driven by the "Composite Man." Instead of forward-scanning for chart patterns, the system performs a brute-force backward sweep across the entire market to find mathematical footprints of supply exhaustion first, before evaluating the structural price context. **Status:** Architecture and Strategy Definition (Day 6 Updates + Edge Cases)

2. Target Universe & Data Boundaries

- **Active Scanning Universe (Tracks A & B):**
 - NIFTY 50
 - NIFTY 100
 - NIFTY MIDCAP 50
 - NIFTY MIDCAP 150
 - NIFTY SMALLCAP 50
 - NIFTY SMALLCAP 250
 - NIFTY MICROCAP 250
 - *Nano-caps included implicitly via threshold logic*
- **Curated Universe (Track C):** Manual static list of 50-100 High-Growth / Sector Leader stocks (avoids the "Denominator Flaw" where retail churn masks delivery percentages).

3. Data Pipeline & Backend

- **Backend Stack:** Kotlin utilizing the `ta4j` library for technical indicator computation.
- **Ingestion Strategy:** Scrape the master NIFTY 500 and NIFTY MICROCAP 250 Bhavcopies + Delivery reports daily.
- **Schedule:** Daily at 18:30 IST.

4. The 3-Phase Pipeline Architecture

The system avoids the "Moving Average Drag" flaw (where sustained institutional buying blinds standard screeners) by utilizing a strict 3-Phase pipeline combining historical density and Z-Scores.

Phase 1: Primary Anomaly Scanners (Backward Sweep)

Scans the universe daily to isolate active institutional footprints using Cap-Adaptive metrics.

- **Historical Volume Shock Anchors (Vectorized via `ta4j`):**
 - *HVQ (Highest Volume in Quarter):* Current volume is the maximum of the rolling 60-day window.
 - *HVY (Highest Volume in Year):* Current volume is the maximum of the rolling 250-day window.

- *HVE (Highest Volume Ever)*: Current volume is the all-time maximum.
- *LVQ (Lowest Volume in Quarter)*: Current volume is the minimum of the rolling 60-day window.
- *LVY (Lowest Volume in Year)*: Current volume is the minimum of the rolling 250-day window.
- *Institutional Bias*: *Is_Blue* (Close > VWAP), *Is_Red* (Close < VWAP).
- **Track A: The Golden Zone (Dual-Confirmation Matrix with Cap-Adaptive Thresholds)**
 - *Cap-Adaptive Delivery Check*:
 - Large/Mid-Caps: > 55%
 - Small-Caps: > 70%
 - Micro-Caps: > 85%
 - Nano-Caps: > 92%
 - *Historical Density*: The Cap-Adaptive threshold must be breached ≥ 4 times within a rolling 15-day window.
 - *Math Shock*: Delivery Volume Z-Score ≥ 2.0 (calculated against a 60-day baseline) to provide double confirmation alongside the rolling density.
 - *Absorption Check*: Daily Price Spread < 20-Day Average Spread (verifies limit order pinning).
- **Track C: The High-Growth Exceptions**
 - *Rule*: Scans the manual 50-100 list exclusively for absolute volume shocks near major structural support, ignoring delivery percentage entirely to bypass the Denominator Flaw.

Phase 2: The 4-Brick Context Filter (The Base Check)

Validates that the anomaly from Phase 1 is occurring in a true Phase A/B "Dead Base" and not a Phase 4 Markdown (falling knife).

- **Brick 1 & 2**: Handled by Phase 1 Delivery/Volume anomalies.
- **Brick 3 (Long-term Gravity)**: *Distance_200DMA* is strictly between -5.0% and +5.0%.
- **Brick 4 (Short-term Stability)**: *ROC_20* sits dead-flat between -5.0% and +5.0% (Sideways Box).

Phase 3: Capital Deployment (Execution Triggers)

Rule: Never buy the raw anomaly to avoid the "Time Trap" (dead capital during Phase B). Capital is deployed strictly via historical volume shock logic:

- **The "Calm Before the Storm" Scanner (Wyckoff Phase C - Spring/Test)**:
 - *Logic*: (*LVY* is True OR *LVQ* is True) AND *ROC_20* is between -5.0% and +5.0%.
 - *Goal*: Catch assets when supply is completely exhausted (absolute zero retail activity), creating the ultimate low-risk floor.
- **The "Institutional Launch" Scanner (Wyckoff Phase D - Breakout/SOS)**:
 - *Logic*: (*HVY* is True OR *HVQ* is True) AND *Is_Blue* is True.
 - *Goal*: Instant execution when momentum algorithms and big money aggressively trigger the markup out of the base.

5. Structural Traps & Edge Cases Mitigated

- **The Distribution Trap (The 52-Week High Flaw)**: Extreme delivery spikes near a 52-week high represent a Buying Climax (institutions distributing into retail FOMO). Phase 2 (Brick 3: 200 DMA

Gravity constraint) mathematically disqualifies these distribution events from being flagged as accumulation.

- **The Denominator Flaw (The NETWEB Exception):** High-growth stocks suffer from massive intraday retail churn, artificially crushing delivery percentages to 20-30% and hiding institutional buying. Track C completely bypasses this by dropping delivery % requirements for the curated list, focusing solely on absolute volume shocks.
- **The "Bad News" Mirage (The Swiggy Scenario):** A stock hitting a 52-week low on bad news paired with a massive delivery spike is the prime footprint of Phase A capitulation. Institutions require this engineered retail panic to vacuum up the float quietly. The pipeline actively targets this setup rather than filtering it out.

6. Execution Rules & Backtesting Framework

- **Signal Generation:** Phase 1 anomalies and Phase 2 context are verified post-market on Day T .
- **Watchlist Staging:** Validated tickers are moved to an active monitoring state awaiting Phase 3 execution triggers.
- **Entry Order:** Triggered on Phase C exhaustion or Phase D launch confirmations.
- **Profit Target/Stop Loss:** To be structurally defined based on asset velocity.
- **Bucket Tracking:**
 - *Closed Bucket:* Target/Stop reached.
 - *Open/Floating Bucket:* Tracking total elapsed holding duration and floating PnL.

7. Open Issues & Pending Decisions

- **Stop-Loss / Take-Profit Architecture:** Structure the exit logic differently for high-velocity assets versus slow-grinding structural assets.

8. Immediate Next Steps

1. Implement custom `ta4j` logic in Kotlin for the Cap-Adaptive Thresholds, 60-day Z-Score, and 15-day rolling frequency logic.
2. Backtest the combined "Track A: Golden Zone" logic on historical data to evaluate the signal-to-noise ratio.
3. Build and manually tag the initial static universe of 50-100 High-Growth / Sector Leader stocks